Experiment 12 - Experiment to detect outliers in a given data set.

Aim:

To handle missing data and inappropriate data in a data set using the Python Pandas library.

Program:

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

np.random.seed(42)
array = np.array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])

print("Array:", array)
print("Mean:", array.mean())
print("25th Percentile (Q1):", np.percentile(array, 25))
print("50th Percentile (Median):", np.percentile(array, 50))
print("75th Percentile (Q3):", np.percentile(array, 75))
print("100th Percentile (Max):", np.percentile(array, 100))

def outDetection(arr):
    Q1, Q3 = np.percentile(arr, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)
print(f"Lower Range: {lr}, Upper Range: {ur}")
sns.displot(array, kde=True)
plt.title("Distribution of Original Array")
plt.show()

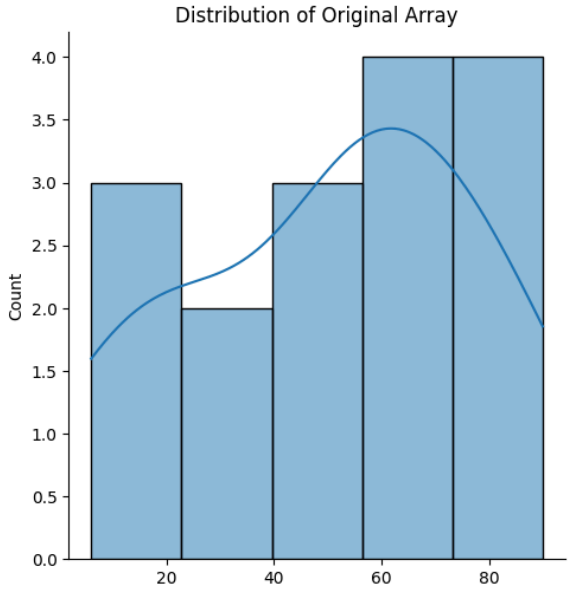
new_array = array[(array >= lr) & (array <= ur)]
print("Filtered Array:", new_array)
lr1, ur1 = outDetection(new_array)
print(f"New Lower Range: {lr1}, New Upper Range: {ur1}")
sns.histplot(new_array, kde=True)
```

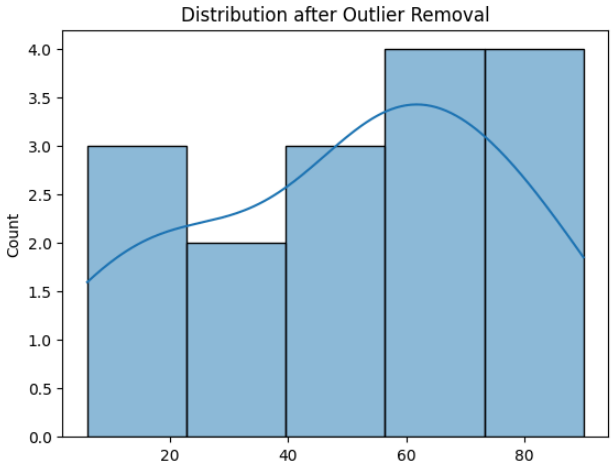
```
plt.title("Distribution after Outlier Removal")
plt.show()
```

Output:

```
Array: [27 50 44  6 58 61 23 86 67 20 75  7 79 61 90 54]
Mean: 50.5
25th Percentile (Q1): 26.0
50th Percentile (Median): 56.0
75th Percentile (Q3): 69.0
100th Percentile (Max): 90.0
```

```
Lower Range: -38.5, Upper Range: 133.5
```





Result:

Thus, the identification of extreme values within the dataset was successfully implemented.